

LANDSBERG AM LECH, Germany

WMO: 108570

Lat: 48.071N

Lon: 10.906E

Elev: 2044

StdP: 13.64

Time Zone: 1.00 (EUC)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.6	13.8	6.6	8.3	9.4	11.4	10.6	15.7	33.7	45.4	29.9	44.2	4.5	170	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	85.6	65.8	82.2	64.8	79.0	63.7	67.3	81.4	65.8	79.2	64.6	77.3	6.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.4	91.0	72.5	60.9	86.1	71.2	59.4	81.6	69.7	32.8	81.9	31.7	79.5	30.7	77.7	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.5	17.1	DB	1.4	90.8	6.1	2.6	-3.0	92.7	-6.6	94.2	-10.0	95.7	-14.5	97.6
			WB	1.1	70.1	6.1	1.2	-3.2	71.0	-6.8	71.7	-10.2	72.3	-14.7	73.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	31.1	32.9	39.8	47.3	55.5	61.8	64.7	64.3	56.3	48.8	39.0	32.9	(6)
	DBStd	14.15	8.78	9.28	7.87	7.38	6.59	6.74	5.94	5.71	6.30	7.30	7.57	8.23	(7)
	HDD50	2542	586	480	323	135	23	2	0	0	14	112	336	531	(8)
	HDD65	6416	1051	900	781	532	303	140	80	83	267	503	780	996	(9)
	CDD50	1798	0	1	8	54	193	355	456	444	205	75	7	0	(10)
	CDD65	190	0	0	0	0	7	43	71	62	7	0	0	0	(11)
	CDH74	2124	0	0	0	10	118	477	750	672	92	5	0	0	(12)
	CDH80	526	0	0	0	1	15	108	211	183	8	0	0	0	(13)
(14) Wind	WSAvg	7.1	8.2	8.3	8.6	7.2	6.8	6.3	6.5	6.0	6.2	6.4	7.2	7.8	(14)
(15) Precipitation	PrecAvg	37.3	2.0	1.9	2.4	2.4	4.1	4.6	4.7	4.3	3.1	2.6	2.6	2.4	(15)
	PrecMax	50.6	4.7	4.0	5.9	5.6	8.4	8.4	9.8	8.7	6.0	6.6	5.9	5.1	(16)
	PrecMin	27.8	0.0	0.6	0.7	0.4	0.3	2.0	0.8	1.6	0.9	0.6	0.0	0.1	(17)
	PrecStd	5.2	1.2	0.9	1.2	1.3	1.9	1.7	1.9	1.9	1.4	1.3	1.4	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	59.1	66.1	75.0	82.3	87.4	90.1	89.8	80.8	73.1	63.0	54.3	(19)
		MCWB	45.8	46.1	50.0	55.9	62.3	66.3	67.0	66.3	63.3	58.7	52.4	45.4	(20)
	2%	DB	50.3	53.6	60.9	70.1	77.3	83.3	85.6	84.6	76.8	68.2	57.6	50.2	(21)
		MCWB	43.8	44.1	48.3	54.0	61.0	65.6	65.9	65.8	62.1	57.5	50.0	44.2	(22)
	5%	DB	47.1	49.3	56.8	66.1	73.2	79.7	81.7	80.9	72.8	64.4	53.6	47.5	(23)
		MCWB	41.6	42.2	46.9	52.6	58.9	64.2	64.8	64.4	60.5	55.3	47.6	42.8	(24)
	10%	DB	43.4	45.9	52.3	61.8	69.4	76.0	78.1	77.3	68.3	60.5	50.0	44.5	(25)
		MCWB	39.5	40.5	44.7	50.3	57.0	62.5	63.7	63.5	58.8	54.0	45.5	40.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.6	47.9	51.9	57.3	64.5	69.0	68.9	68.6	64.8	60.4	53.1	47.3	(27)
		MCDB	52.8	56.4	62.4	71.6	78.2	82.3	84.7	83.2	77.9	69.9	60.4	52.4	(28)
	2%	WB	44.5	44.8	49.4	54.9	62.0	66.6	67.1	66.9	63.0	58.3	50.6	45.3	(29)
		MCDB	49.5	51.7	58.5	67.8	75.1	80.1	81.4	80.8	74.1	66.2	56.5	49.5	(30)
	5%	WB	42.0	42.6	47.4	53.0	59.8	65.0	65.7	65.5	61.2	56.3	48.4	43.1	(31)
		MCDB	46.2	48.6	55.3	64.4	71.0	78.1	78.9	78.6	70.9	62.9	53.0	46.7	(32)
	10%	WB	39.6	40.3	45.4	50.9	57.7	63.0	64.3	64.0	59.4	54.3	46.2	40.8	(33)
		MCDB	43.1	45.3	51.9	60.6	67.5	74.4	76.5	76.2	67.0	59.7	49.6	43.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	14.0	16.3	18.9	18.8	18.7	19.2	19.4	17.4	15.8	12.3	11.2	(35)
		MCDBR	16.3	20.7	24.7	27.6	26.9	25.9	26.8	26.7	25.4	22.4	19.7	16.1	(36)
	5% WB	MCWBR	11.0	13.0	14.3	14.3	13.2	11.9	11.1	11.5	13.0	13.0	12.9	10.9	(37)
		MCWBR	15.1	18.8	22.2	25.2	24.0	23.6	23.7	24.2	22.5	20.1	17.7	14.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.286	0.310	0.344	0.377	0.388	0.393	0.395	0.389	0.365	0.336	0.308	0.283	(40)
		taud	2.459	2.396	2.368	2.301	2.308	2.324	2.310	2.345	2.392	2.466	2.498	2.475	(41)
	Ebn at Noon		254	270	277	277	278	276	273	270	266	257	243	240	(42)
		Edh at Noon	20	27	32	38	39	39	39	36	31	25	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		355	636	988	1361	1564	1752	1698	1493	1085	702	384	289	(44)
	RadStd		46	84	119	180	180	160	157	153	132	82	52	38	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (87 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
						+0.74	+0.72	N/A	N/A	+106	+36

Nomenclature: See separate page